Green Team P1: Brainstorming Presentation

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Overview

- Team Background & Schedule
- Task Interpretation & Goals
- Brainstorming & Article
- Hardware Improvements
- Design 1: Kalman Filter + Bang Bang
- Design 2: Particle Filter + P Steering
- Expected failure points

Team Background

Ben Bruick	Layla Chehab	Aidan Deacon	Kimberly Gurwin
Robotics BSE	Electrical BSE	Robotics BSE	Robotics BSE
Software & Hardware Design	Software Development	Hardware Design & Manufacturing	Hardware Design & Manufacturing

Team Schedule

- **Class:** Tue/Thu 12:00-2:30 pm (Sep 30, Oct 2, Oct 7, Oct 9, Oct 14, Oct 16).
- **Team work time (after class):** 2:30-4:30 pm (Sep 30, Oct 2, Oct 7, Oct 9, Oct 14, Oct 16).
- **Sunday sessions:** 2:00-5:00 pm (Sep 28, Oct 5, Oct 12).
- **Milestones:**
    - **Sep 23** Consultation
    - **Sep 25** Brainstorming Presentation
    - **Oct 7** Demo 1
    - **Oct 16** Demo 2
    - **Oct 21** P-Day

Task Interpretation - Demos

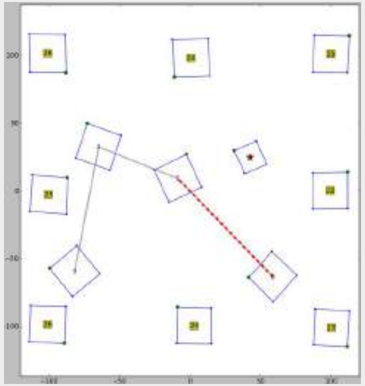
Demo Day 1

- *RobotSimulatorApp* runs successfully
- Responds to user commands
- Displays and updates robot state information graphically
- Updates estimate with robot motion

Demo Day 2

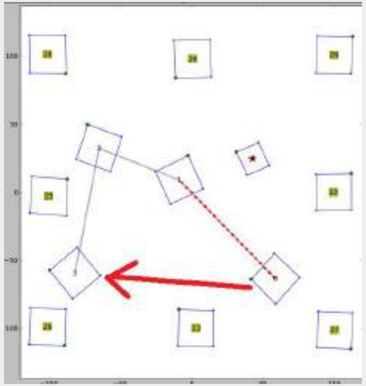
- Robot responds to manual commands
- 90° steering/turning/left or right motion capability
- Switches to autonomous mode
- Robot moves between at least two waypoints (Auto. mode)

Task Interpretation: P-Day



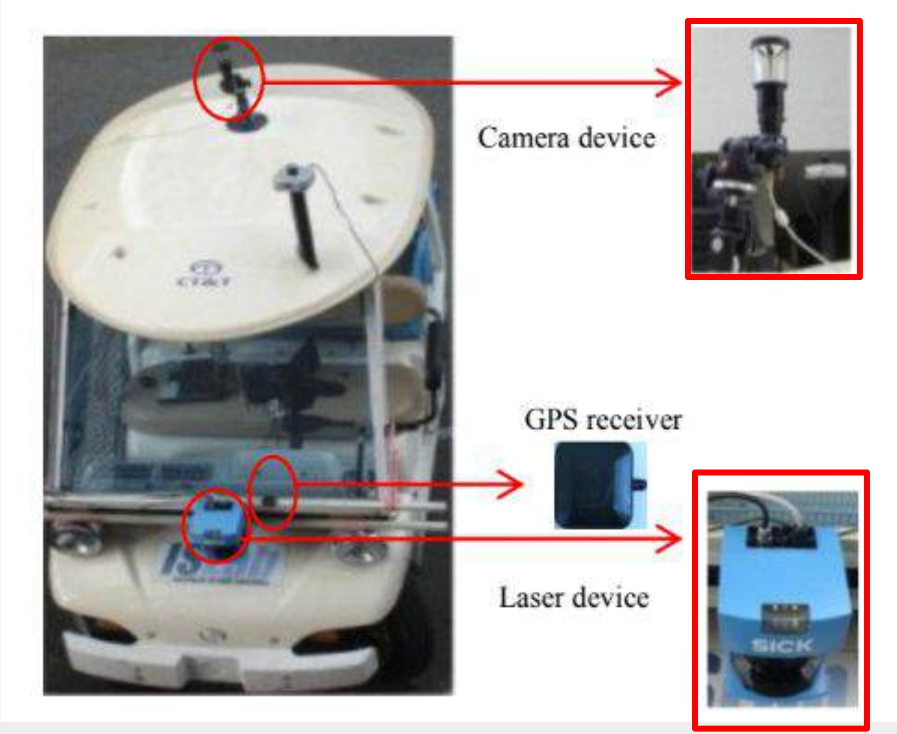
Competition:
- Robot navigates between a sequence of waypoints autonomously in the shortest possible time.
- Robot avoids leaving the designated arena, which is constructed using 8 tags.

Pass-Fail:
- Robot navigates to the final waypoint in under 15 minutes.
- Robot tag remains 1m above the floor.
- Robot does not leave the arena.

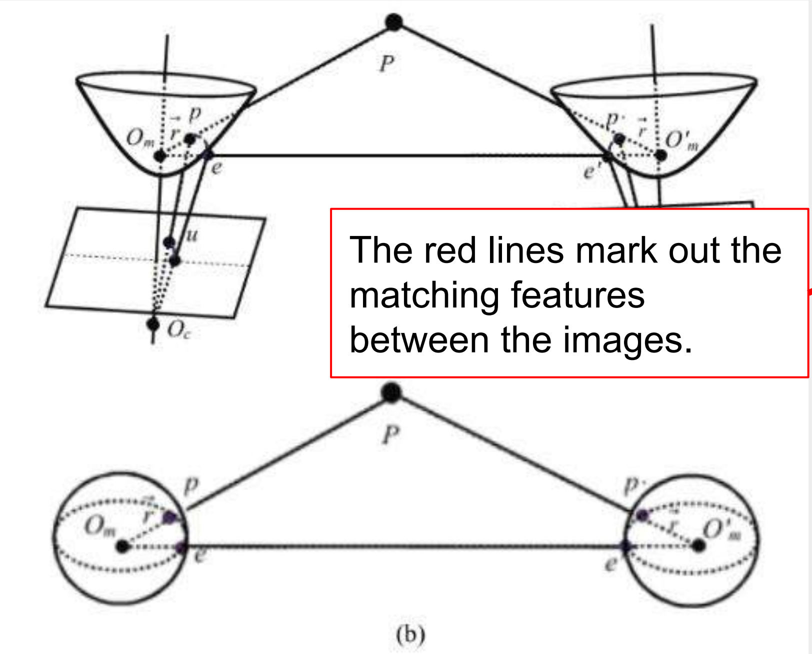


Academic Article

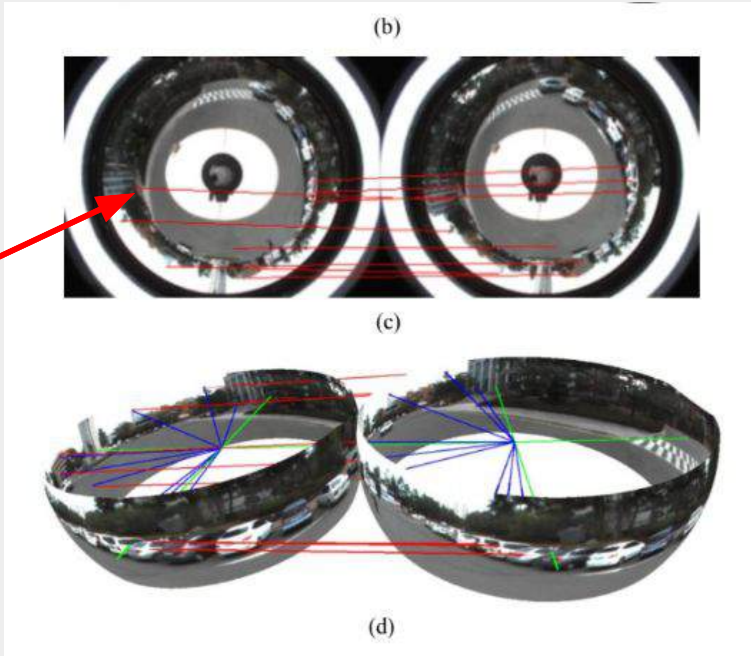
The University of Ulsan in South Korea used an extended Kalman filter (EKF) to predict the location of their autonomous vehicle.



The camera they used was 360° views, which meant they had to adjust the warping of the image.



The red lines mark out the matching features between the images.

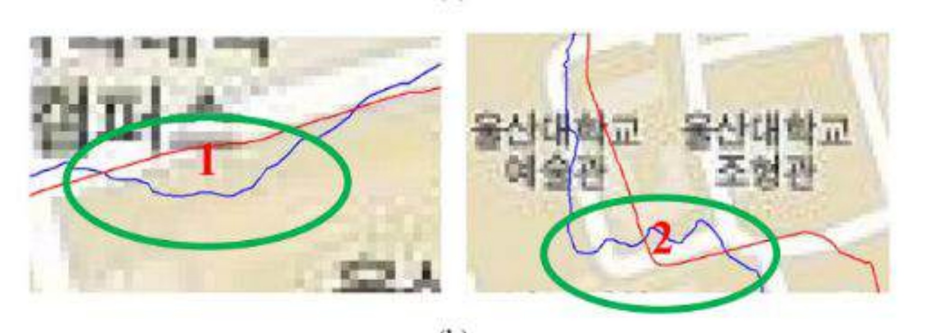


State variables were defined from the position of the vehicle and the landmark.

$$\mathrm{V} = \begin{pmatrix} x_v\text{: vehicle location in x} \\ y_v\text{: vehicle location in y} \\ z_v\text{: vehicle location in z} \\ \alpha_v\text{: vehicle location in } \alpha \\ \beta_v\text{: vehicle location in } \beta \\ \varphi_v\text{: vehicle location in } \varphi \end{pmatrix}$$

$$\mathrm{L} = \begin{pmatrix} x_l\text{: landmark location in x} \\ y_l\text{: landmark location in y} \\ z_l\text{: landmark location in z} \end{pmatrix}$$

EKF improves localization through noisy signal readings.



(b)

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This table uses the GPS as the baseline measurement for how accurate each model is to it's expected endpoint.

TABLE II. COMPARISON GPS DATA, LOCALIZATION USING EKF AND WITHOUT USING EKF

	At the end of travel			Maximum difference to GPS (meter)
	Latitude	Longitude	Error (meter)	
Our method	35.54466	129.25683	*74.5*	*70.8*
V.O. without EKF	35.54402	129.25691	*95.4*	386.2
GPS	35.54445	129.25625	3.8	

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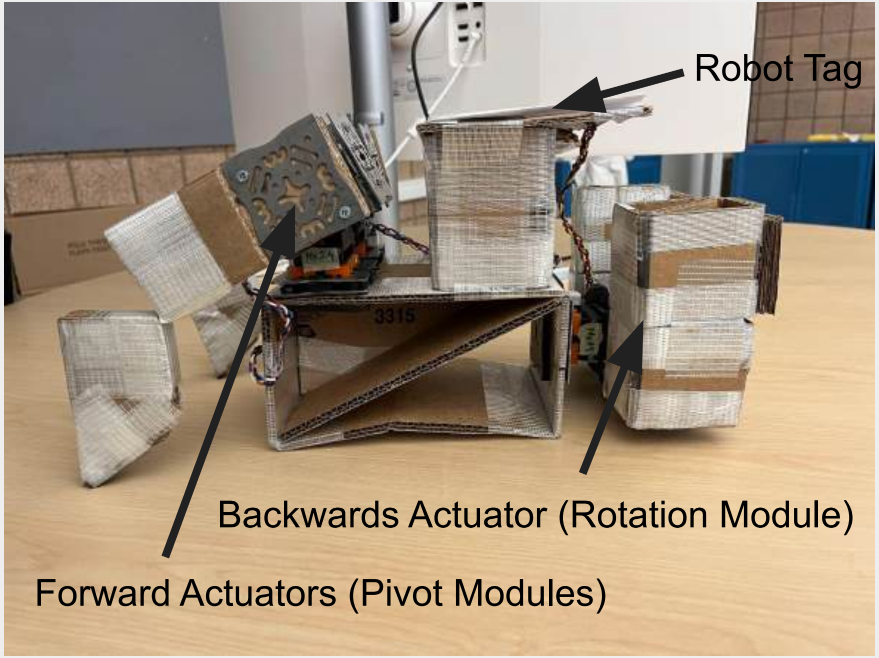
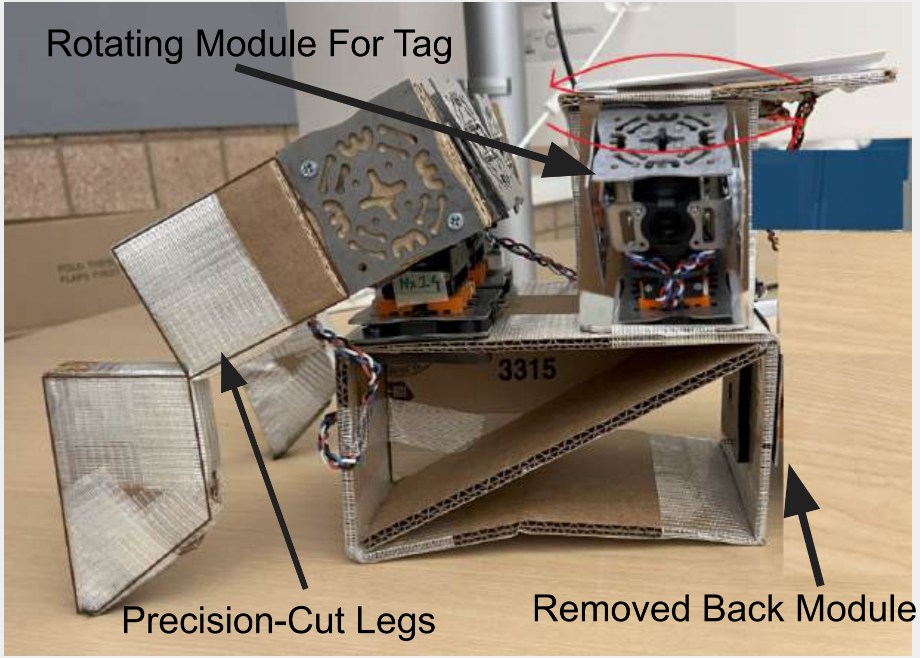
Hardware Improvements

Current Design:



Modifications:



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Design 1:
Kalman Filter + Bang-Bang

State

- Cross track error
- Heading error vs segment
- Dist to waypoint
- Waypoint list

Actions

- Steer left, steer right, or go straight
- Move forward at slow or fast speed
- Stop

State Estimation

- **Initialization**: First AprilTag pose gives cross-track and heading error
- **Prediction**: Kalman Filter uses commanded forward speed and turn rate
- **Update**
  - Virtual front and back sensor distances
  - Spinning robot tag
- **Dominant errors**: Sensor saturation, parallax from tag height, inconsistent movement

Policy

- Stop if waypoint reached, steer toward next expected heading
- Steer inward and slow down if boundary warning
- Otherwise, bang-bang rules on heading error
  - Far left = Steer right @ slow speed
  - Far right = Steer left @ slow speed
  - Balanced = Straight @ fast speed
- Hysteresis prevents constant switching

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Design 2: Particle Filter + P Steering

State

- Cross-track error and heading error
- Distance to next waypoint
- Desired heading from Pure Pursuit lookahead point
- Waypoint list

Actions

- Continuous steering angle within safe range
- Continuous forward speed between stop and max
- Stop

State Estimation

- **Initialization**: Spread particles around first AprilTag pose (W0)
- **Prediction**: Particle Filter propagates particles using commanded motion
- **Update**:
  - Predicted vs measured sensor distances
  - Spin to refine relative heading
- **Pure Pursuit**: Computes desired heading from lookahead point
- **Dominant errors**: Sensor parallax, lookahead distance too short or long

Policy

- Stop if waypoint reached or data is stale
- Steer inward and slow if boundary warning
- Else
  - Compute heading error to Pure Pursuit target
  - Steering angle set by proportional control on heading error
  - Forward speed fast when aligned and far, slow when close or misaligned

Quantitative Analysis

- Base path: 3.0 m (3x 1 m legs)
- **P0 Straight line speed**: 1 m / 47.42 s = 0.0211 m/s
- Path stretch: **Bang-Bang** +10% = 3.3 m; **Pure Pursuit** +2% = 3.06 m
- **Bang-Bang speeds**: 60% at 0.0211 m/s, 40% at 0.0101 m/s = avg 0.0167 m/s
- **Pure Pursuit speeds**: 80% at 0.0211 m/s, 20% at 0.0127 m/s = avg 0.0194 m/s
- **Bang-Bang travel time**: 3.3 / 0.0167 = 197.6 s
- **Pure Pursuit travel time**: 3.06 / 0.0194 = 157.7 s
- **Total Time:** Bang-Bang = 199.6 s total
- **Total Time:** Pure Pursuit = 158.7 s total
- **Net: Pure Pursuit ~20% faster**

Expected Pain Points

- Noisy / clipped sensor values
- Ambiguity in heading estimation from two perpendicular sensors
- Risk of drifting outside the 2 m x 2 m arena
- Controller oscillation from bang-bang / poorly tuned P-cont.
- Parallax error with robot tag

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Thank You

Resources

[1] V.-D. Hoang, M.-H. Le, D. C. Hernández, and K.-H. Jo, “Research guides: Research impact metrics: Citation analysis: Scopus,” Scopus - Research Impact Metrics: Citation Analysis - Research Guides at University of Michigan Library, https://guides.lib.umich.edu/citation/Scopus (accessed Sep. 24, 2025).

Extended Kalman Filter

Extended Kalman Filter

Tested against other filters & GPS

It validates the effort necessary to use a predictive model.

It can reduce the error from noisy measurements – which we know because noisy stuff is provided.

Map w/ 3 lines -> filter improves the quality of the localization.

Not concerned with the intrinsic properties of the camera or laser or camera/stuff for me.

We can mention camera distortion stuff especially because the tag is mounted on top of the robot in the same way that the camera was mounted on top of the car.

https://www-scopus-com.proxy.lib.umich.edu/pages/publications/84893536601?origin=resultslist